

High-Dimensional Latent Dynamics and Structural Mapping:

*An Analysis of How Generative Transformers Parse Prompt
Architectures*

Latent Space Geometry and Tokenization Mechanics

Source guide

This technical analysis explores how generative transformers transform textual inputs into dynamic latent representations through a sophisticated pipeline of tokenization, embedding, and self-attention. The text details the evolution of positional encoding systems, such as RoPE and ALiBi, which are essential for helping models understand sequence order and scale to long-context windows. Beyond basic mechanics, the source examines the mathematical decay of attention influence and the efficiency of various structural scaffolding formats like JSON or Markdown in prompt architecture. A significant portion of the work focuses on post-training alignment through techniques like RLHF and DPO, while simultaneously addressing security vulnerabilities like prompt injection and jailbreaking. Ultimately, the analysis provides a comprehensive framework for understanding how modern AI balances semantic reasoning, computational efficiency, and robust adversarial defense within high-dimensional spaces.

Generative artificial intelligence models process user prompts through a systematic translation of textual instructions into **high-dimensional vector representations** that traverse multi-layered neural networks.[1, 2]

This mapping process begins at the **tokenization boundary**, where an input sequence is divided into discrete linguistic units known as **tokens**. [3, 4] These tokens correspond to words, sub-words, or character sequences, and are assigned unique integer **token IDs** representing their coordinates within a pre-defined **vocabulary lookup table**. [3, 4]

emb

∈ R
d
model

$$W_{emb} \in R^{d_{model} \times V}$$

To initiate mathematical computation, each token ID is converted into a **V-dimensional one-hot encoded vector**, where V represents the total vocabulary size.[3, 4]

This vector is projected onto a learned **embedding matrix** to extract a dense, continuous starting representation of dimension d_{model} . [3, 4]

model

While early architectures relied on **static embeddings** that mapped homonyms or polysemous words to a single compromised coordinate, modern transformers generate **dynamic contextual embeddings**. [3, 4, 5]

The **self-attention mechanism** updates these initial vectors by shifting their coordinates in the multi-dimensional latent space based on surrounding tokens, clustering semantically related concepts dynamically.[4, 6]

This contextualization is not restricted to text; modern transformer backbones apply uniform tokenization and embedding principles across diverse prompt modalities, representing protein sequences, image patches, and video frames as sequence matrices $X \in \mathbb{R}^{N \times d}$

$N \times d$
model

[1, 7, 8]

The mathematical engine driving this contextual translation is the **scaled dot-product attention block**. [4, 6]

For an input matrix X , the model applies learnable weight projections to generate **Query (Q)**, **Key (K)**, and **Value (V)** representations [3, 4, 6]:

$$Q = XW_Q, K = XW_K, V = XW_V$$

The relationship between these matrices determines how tokens prioritize contextual cues, calculated as [4, 6]:

$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$



QK^T
T

$\frac{1}{\sqrt{d_k}}$
The scaling factor
 d_k
k



normalizes the dot product to preserve gradient stability during optimization.[4, 6]

From a **dynamical systems perspective**, the layer-wise evolution of these token representations can be modeled as a **mean-field interacting particle system**.^[7]

Here, individual tokens behave as physical particles whose trajectories across successive layers are governed by their **empirical measure**, which represents the spatial distribution of the token point cloud in latent space.^[7]

As tokens propagate through deep networks, this particle system tends to exhibit clustering behavior in the later layers. This spatial convergence is mathematically associated with the phenomenon of **rank collapse**, where the dimensionality of the token representations diminishes, leading to highly correlated vectors.^[7]

Mechanistic interpretability reveals a structural correlation between the geometric properties of these token representations and model performance.^[7]

Specifically, prompts that incur higher average **cross-entropy loss** project their token representations into higher-dimensional intrinsic spaces across the hidden layers, while predictable, low-entropy prompts collapse tokens onto lower-dimensional manifolds.^[7]

To improve alignment, techniques like **Syntax-Semantic Contrastive Pretraining (SSCP)** employ dual-encoder frameworks to align standard contextual embeddings with graph-based positional encodings derived from dependency trees, forcing the shared embedding space to represent both structural syntax and semantic views.^[9]

However, standard self-attention blocks are prone to inherent **ill-conditioning** during optimization, where the condition number of the self-attention matrix closely scales with that of the embedded token matrix.^[8]

This ill-conditioning slows down gradient convergence and introduces training instability.^[8] To mitigate this, models can apply **conditioned embedded tokens**, which introduce a structured correction term to the first-layer embedding matrix to reduce its condition number.^[8]

For massive sequence mapping, linear approximations like the **Nystromformer** substitute standard quadratic complexity with near-linear approximations, preserving global dependency modeling while reducing the computational footprint.^[8]

Positional Encoding Systems and Structural Mismatches

Because the scaled dot-product attention calculation is permutation-invariant, transformers possess no inherent understanding of sequence order.^[1, 10] Without explicit positional signals, the attention mechanism is unable to distinguish structural or semantic variations caused solely by word order.^[10, 11] Positional encoding systems resolve this limitation by injecting spatial coordinates into the token representations.^[10, 11]

These spatial encoding techniques have evolved across three developmental eras: absolute static additions, relative scalar biases, and rotational complex projections.^[11] Modern architectures largely favor relative positional frameworks, which prioritize the distance between tokens over their absolute coordinates.^[10, 11, 12]

Paradigm	Core Mechanism	Advantages	Constraints
Absolute Learned Embeddings [10, 11]	Adds coordinate vector to embedding: $X = V + P$.	Simple implementation.	Limited to training context length.
Rotary Position Embedding (RoPE) [12, 15, 16]	Rotates queries/keys in complex plane.	Robust for deep architectures; explicit relative distance.	Increases decoding overhead.

When scaling large language models to process long context windows, standard positional encodings generate out-of-distribution (OOD) values at extended token positions, causing catastrophic retrieval failures.[15, 18, 19] To resolve this, several scaling strategies have been developed:

- Positional Interpolation (PI): Linearly scales positional indices to compress the extended context back into the pre-trained limits.[15, 16]
- NTK-Aware Scaling: Rescales the base frequencies of RoPE to interpolate low-frequency dimensions while preserving high-frequency details.[15, 16]
- YaRN (Yet another RoPE extensionN): Extrapolates context by dividing hidden dimensions into distinct frequency bands, adjusting only those dimensions that receive inadequate training.[15, 18]
- LongRoPE2: Focuses on the unequal training of RoPE dimensions, where higher dimensions—critical for long-context processing—remain under-trained.[19] LongRoPE2 employs an evolutionary search guided by "needle-driven" perplexity, which measures the retrieval of specific answer tokens hidden deep within long documents.[19] To preserve short-context performance, it applies mixed context window training, packing short and long sequences together while applying attention masks to prevent cross-document leakage.[19]
- Clipped RoPE (CoPE): Introduces a soft clipping strategy to stabilize the rotation angles over long distances, offering a scalable solution for long contexts.[12]
- Mesa-Extrapolation: Uses a chunk-based triangular attention matrix and applies Stair PE to manage the final chunk, reducing memory overhead and accelerating inference.[13]

These positional frameworks often encounter structural mismatches in real-world translation tasks, such as Simultaneous Machine Translation (SimulMT).[17] In SimulMT, the incremental, streaming arrival of source tokens continuously shifts the absolute positional indices of subsequent tokens.[17] This positional shift causes a mismatch with the cached Key-Value (KV) states of earlier segments.[17] Recomputing the KV cache restores positional consistency but incurs prohibitive computational overhead, creating a direct conflict between inference efficiency and positional consistency.[17]

While Group Position Encoding (GPE) attempts to resolve this by assigning independent positional groups to source and target segments, its fractional or zero intervals can violate standard pre-training objectives.[17]

Similarly, hierarchical models resolve context boundaries by partitioning tokens into local utterances (using local positional encodings and block-diagonal UT-masks) and global dialog-level contexts (using global positional encodings and CT-masks), allowing selective cross-utterance attention without losing boundary clarity.[20]

Mathematical Limits of Causal Masking and Attention Decay

Autoregressive transformer decoders rely on causal attention masking to prevent tokens from attending to future positions during generation.[14, 21, 22] While this causal structure ensures sequential generation, it imposes strict mathematical constraints on the propagation of gradients and contextual information across the sequence.[14]

Theoretical investigations reveal that causal masking, when combined with standard residual connections, forces a mathematical polarization of attention influence.[14] This behavior is defined by two distinct phenomena:

Primacy Tail~ $O(\log N)$, Recency Delta~ $O(1)$

The causal mask forces a logarithmic divergence of gradient influence at the start of the prompt, creating the Primacy Tail.[14] Conversely, residual connections preserve an isolated $O(1)$ anchor at the final token, creating the Recency Delta.[14]

Between these two extremes lies a factorial dead zone of order [14]:

$$O\left(\frac{(H-1)!}{1}\right)$$

where H represents the number of attention heads. Inside this dead zone, the model's ability to maintain state transitions decays rapidly.[14] This mathematical decay limits the transformer's capacity to maintain stack-like behavior, preventing it from tracking and resolving multiple levels of nested recursive dependencies over long sequences.[14] This decay also contributes directly to the U-shaped "Lost in the Middle" retrieval curve, where models struggle to retrieve information positioned in the middle of long contexts.[14]

Beyond length-based decay, standard attention mechanisms are prone to the attention skew effect when processing structured histories, such as user interaction sequences in recommendation systems.[23, 24] Because tokens belonging to the same item are positionally closer and co-occur frequently in training data, the self-attention mechanism disproportionately focuses on these intra-item tokens.[23, 24] This local focus prevents the model from capturing broader cross-item correlations.[23, 24]

To counter this skew, Hierarchical Attention Masking (HatLLM) dynamically modifies the attention mask across different layers of the model.[23, 24]

Layer Depth	Masking Strategy	Target Semantic Operation
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Shallow Layers [23, 24]	Masks attention between tokens of different items.[23, 24]	Facilitates clean, isolated intra-item semantic understanding.[23, 24]
Middle Layers [23, 24]	Retains the original unmasked attention mechanism.[23, 24]	Preserves fine-grained local and global token relations.[23, 24]
Deep Layers [23, 24]	Masks attention within individual items.[23, 24]	Forces the model to focus on and capture long-range cross-item correlations.[23, 24]

By implementing this layer-wise masking structure, models achieve significant performance improvements, demonstrating an average retrieval accuracy gain of 9.13% over standard causal masking baselines across real-world datasets.[23, 24]

Structural Scaffolding and Parsing Efficiency

Generative language models do not parse structured formats like JSON, XML, YAML, or Markdown as native, schema-validated objects.[2] Instead, the tokenizer reads every structural character—such as repeated keys, brackets, and whitespace—as individual tokens.[2] Consequently, heavy structural scaffolding consumes valuable context window space and increases billable token costs.[2]

This token inflation can crowd out instructions and source documents, leading to attention dispersion and degrading model performance on complex downstream tasks.[2, 25] To optimize context efficiency and output integrity, developers must evaluate structured prompt architectures on their ability to compress data while preserving structural and semantic relationships.[2]

Structured Prompt Format	Token Efficiency Profile	Structural and Schema Integrity	Production Context and Mechanics
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Lossless Evidence Aliases [2]	Excellent (~75% reduction in input token size).[2]	Very High (retains stable references for validation).[2]	Normalized reference format. Declares repeated metadata and source files once with short unique IDs (e.g., F001, S001), referencing them in compact data rows.[2]
TOON-Style Tables [2]	High (eliminates repeated object keys in arrays).[2]	High (explicit row mapping).[2]	Token-Oriented Object Notation. Declares schemas once at the header, then streams structured rows.[2]
CSV / TSV with Metadata [2]	High (compact tabular layout).[2]	Moderate (struggles with deeply nested objects).[2]	Tabular format. Integrates flat tables with explicit metadata blocks defining null values and unit rules.[2]
XML-Tagged Sections [2, 26]	Low (introduces structural token overhead).[2]	Excellent (clearly demarcates prompt boundaries).[2]	Semantic division. Clearly separates system instructions, few-shot examples, and source documents.[2]
YAML [2]	Moderate (removes some bracket overhead).[2]	Moderate (highly sensitive to indentation errors).[2]	Configuration format. Best reserved for human debugging and prompt drafting.[2]

Minified JSON [2]	Low (only removes whitespace).[2]	High (standard machine-readable format).[2]	Serialization format. Useful for small, highly irregular, or nested structures where flattening is impractical.[2]
Pretty JSON [2]	Very Poor (massive whitespace and key bloating).[2]	High (standard format).[2]	Debugging format. Highly inefficient for production pipelines due to excessive token consumption.[2]

To maintain efficiency and reliability in production systems, prompt engineering frameworks distinguish between the representation used for inputs and outputs.[2] While input data is compressed using token-efficient structures like lossless evidence aliases, output formats are strictly enforced using constrained decoding.[2, 27]

If models are guided solely by prompt instructions (e.g., "Return a valid JSON object"), they frequently fail on edge cases, return conversational preambles, or wrap their responses in markdown code blocks.[27] Providers implement native structured outputs by applying a mathematical mask over the model's vocabulary during token generation.[27] Guided by a Finite State Machine (FSM) that tracks the exact state of a predefined JSON schema, the constraint layer sets the probability of all syntactically invalid tokens to zero.[27] This mathematically forces the model to select a schema-compliant token.[27]

However, systematic evaluations from LLMStructBench, JSONSchemaBench, and StructEval reveal a notable trade-off when using these constraints.[28] Enforcing rigid grammar-constrained decoding can distort the model's token probability distribution, leading to syntactically valid but semantically degraded outputs that reduce the model's reasoning capabilities.[28]

To balance validation with performance, developers utilize specialized output parsers (such as LangChain's JsonOutputParser) to stream and parse partial JSON objects, alongside Markdown layouts that leverage headings and bulleted lists to organize complex contexts naturally.[25, 26]

In-Context Prompt Engineering Paradigms and Tool Augmentation

In-context learning (ICL) is an emergent capability of large language models, allowing them to perform complex tasks by conditioning their predictions on a set of demonstrations without updating their weights.[29, 30, 31]

This paradigm encompasses several key prompt architectures, including zero-shot learning (direct task instructions), few-shot learning (providing high-quality input-output demonstrations), and Chain-of-Thought (CoT) prompting (generating step-by-step intermediate reasoning steps to solve complex mathematical or logical tasks).[29, 30, 31, 32] To improve reasoning stability, self-consistency sampling generates multiple outputs with a temperature higher than zero, selecting the final answer via a majority vote.[32]

However, few-shot CoT prompting can introduce semantic noise and distracting tokens that decay performance.[33, 34] To mitigate this, Few-shot Attention Intervention (FAI) dynamically analyzes attention patterns over the demonstrations to identify distracting tokens, applying a targeted correction to suppress their weight.[33]

Additionally, unsupervised exemplar selection often lacks scalability and introduces irrelevant semantic noise.[34] To improve performance, models use nearest-neighbor (k-NN) clustering in the embedding space to select exemplars that are semantically similar to the target query.[32]

Advanced logical frameworks extend these paradigms to structured reasoning:

- Chain-of-Thought with Explicit Evidence Reasoning (CoT-ER): Divides relation extraction into a three-step reasoning process, requiring the model to identify concept-level head and tail entities before extracting relevant contextual spans as explicit evidence.[30]
- Temporal Action Localization (TAL): Applies hierarchical reasoning to video and text, guiding models to identify temporal dependencies and causal relationships to align video segments with textual actions.[35]

To execute actions in external environments, models employ tool-augmented architectures.[32]

Tool Augmented Language Models (TALM) generate specialized tokens (such as [tool-call]) to trigger external API calls, appending the returned results directly back into the sequence context.[32]

Similarly, Toolformer is trained in a self-supervised manner to autonomously decide which tool to call, what arguments to pass, and when to integrate the returned API response.[32]

Input: "The capital of France is Paris."

Toolformer Annotation



"The capital of France is [QA(France) → Paris]."

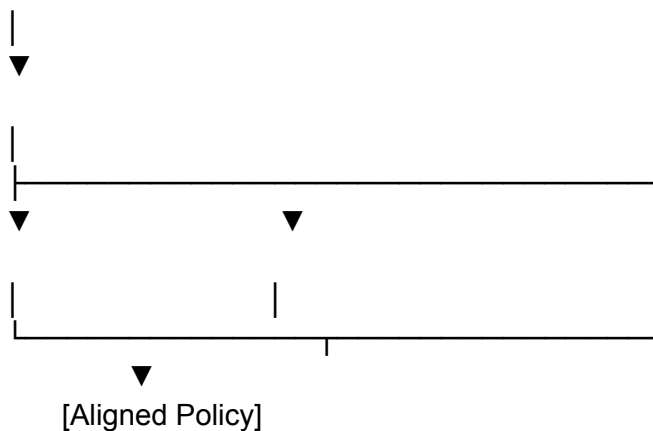
As language models scale, the utility of in-context learning has shifted.[31] For advanced models like the Qwen2.5 series, evaluations show that adding traditional few-shot CoT exemplars does not improve reasoning performance compared to zero-shot CoT prompting.[31]

These highly capable models tend to bypass the detailed logical content of exemplars, focusing directly on the final instructions.[31] Instead, the primary function of few-shot demonstrations in advanced models is formatting alignment—guiding the model to match the specific output layout expected by the user.[31]

Post-Training Alignment and Intent Translation

While pre-training compiles broad linguistic knowledge, it optimizes models to minimize token prediction error on massive text corpora rather than to follow instructions.[36, 37, 38] This mismatch often results in verbose, chaotic, or unsafe outputs.[37]

To align models with human intentions, safety guidelines, and formatting standards, they undergo a series of post-training alignment stages.[37, 38, 39]



Supervised Fine-Tuning and Instruction Tuning

Supervised Fine-Tuning (SFT) trains pre-trained models on curated datasets of instruction-response pairs.[36, 37, 39] This process adapts model parameters using gradient descent to minimize next-token prediction error on expected completions.[37, 38]

To build diverse instruction datasets, frameworks utilize expert annotation, self-improvement loops, or open-source templates like the Public Pool of Prompts (P3).[36, 39] P3 maps conventional natural language processing tasks into structured instruction datasets, utilizing PromptSource templates to format inputs and expected outputs.[36]

During SFT, models are trained on the entire prompt-completion sequence, which can cause them to overfit to prompt phrasing.[40] SFT trainers address this by masking prompt tokens using the `ignore_index=-100` parameter in PyTorch's `CrossEntropyLoss`, calculating gradients exclusively on completion tokens.[40]

To balance instruction tuning with formatting flexibility, trainers can replace this binary masking switch with a continuous Prompt-Loss-Weight (PLW) parameter, where modulating $0 < \text{PLW} < 1$ dampens the gradient influence of prompt tokens during backpropagation.[40]

Preference Optimization Frameworks

While SFT aligns models with task formats, it struggle to resolve subjective preferences such as helpfulness and safety.[37, 38] To address this, models undergo preference alignment.[37, 41, 42]

Under the standard Reinforcement Learning from Human Feedback (RLHF) pipeline, human annotators rank multiple model completions for a single prompt x , training a parameterized reward model r

ψ

(x, y) to assign higher scores to preferred completions y

w

than to dispreferred ones y

I

[41, 42] This optimization uses a sigmoid loss function [42]:

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$(x, y$

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$(x, y$

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$(x, y$

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$))]$

Once the reward model is trained, the policy π

θ

(the LLM) is optimized to maximize the expected reward while applying a Kullback-Leibler (KL) penalty to prevent the model from drifting too far from the reference SFT policy π_{ref}

[37, 42]:

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 $\max$

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 $x \sim D, y \sim \pi$
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$(x, y)] - \beta D$
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Direct Preference Optimization (DPO) simplifies this process by bypassing the reward model and the unstable online reinforcement learning loop.[36, 37, 42] DPO derives an analytical relationship showing that the reward model can be expressed directly in terms of the policy π_{θ}

and reference policy π_{ref}

. [42]

The model is optimized directly on preference pairs [37, 42]:

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However, applying negative weights to dispreferred responses in DPO can introduce training instabilities because the loss landscape is unbounded.[42]

To stabilize alignment, variational offline frameworks like Advantage Leftover Launch (A-LoL) reformulate the optimization as a reward-driven re-weighted SFT loss, utilizing exclusively preferred responses to align the model.[42]

Additionally, models can employ contextual fine-tuning, prepending structured prompt contexts directly to raw training sequences to guide domain-specific learning.[43]

Adversarial Prompt Architectures and Safeguards

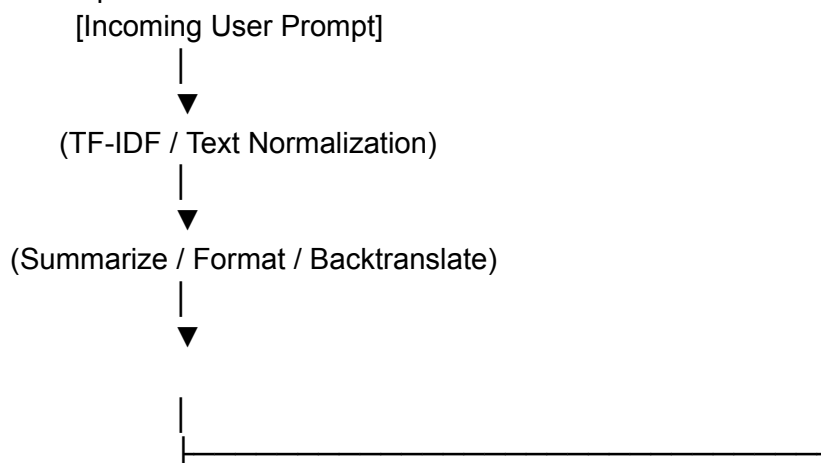
The open-ended natural language interface of generative language models introduces significant security vulnerabilities, primarily prompt injection and jailbreaking.[44, 45, 46]

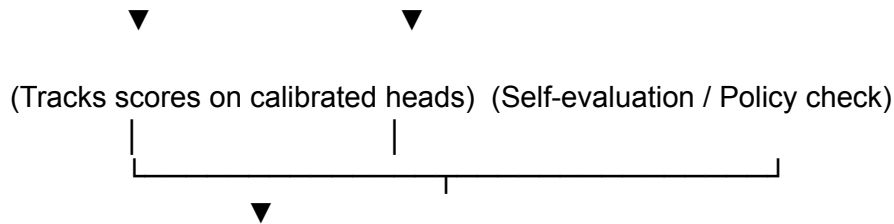
These exploits manipulate the model's text-prediction objectives to bypass safety guidelines [45, 47, 48]:

- Jailbreaking: Adversarially engineered inputs designed to suppress the refusal responses learned during post-training alignment.[47, 48] Common tactics include roleplay overrides (e.g., instructing the model to act as "DAN" - Do Anything Now), instruction layering (embedding harmful queries within deep fictional scenarios), obfuscation (base64 encoding or multilingual translation), and contradiction forcing (forcing compliance by framing refusal as disobedient).[47, 49]
- Prompt Injection: Direct or indirect manipulation that overrides system-level instructions.[50] Direct injections involve immediate malicious instructions, while indirect injections hide instructions inside external data sources (e.g., resumes, web pages, or retrieved RAG documents) that the AI system processes.[50, 51] A primary risk of these injections is data exfiltration, where the model is tricked into leaking system prompts, API keys, or personal data to unauthorized channels, leading to severe legal and compliance breaches under HIPAA or GDPR.[45, 50]

At the network level, these attacks exploit the distraction effect.[52] Mechanistic evaluations demonstrate that when faced with an adversarial prompt, the attention scores of specific attention heads shift away from the system instructions and focus entirely on the injected malicious tokens.[52]

To counter these vulnerabilities without fine-tuning, several runtime defense frameworks have been developed:





1. Attentive Monitoring: RENNERVATE

The RENNERVATE framework leverages fine-grained attention features at the token level to detect and neutralize indirect prompt injections.[22] It employs a two-step attentive pooling mechanism that aggregates attention scores across attention heads and response tokens during inference.[22] By identifying when attention shifts to untrusted input zones, it allows precise, lightweight sanitization that neutralizes the injection while preserving the model's helpful output.[22]

Similarly, runtime attention tracking monitors key attention heads—calibrated beforehand using random sentences—by measuring attention scores from the last token back to the system prompt, flagging sudden score drops as potential injections.[52]

2. In-Context Adversarial Game (ICAG)

ICAG implements an iterative adversarial game between an attack agent and a defense agent.[53] The attack agent analyzes why previous jailbreak attempts failed and refines its tactics.[53] Meanwhile, the defense agent reflects on successful and failed jailbreak attempts, extracting defense patterns to construct dynamic, highly resilient system instructions.[53] This continuous defensive adaptation occurs purely in-context, without requiring expensive retraining.[53]

3. Semantic-Preserving Transformations and Backtranslation

Adversarial jailbreak prompts are often highly fragile, relying on specific word arrangements to trigger the alignment bypass.[54] Semantic smoothing applies transformations to neutralize these attacks [54, 55]:

- Summarization: Condenses the input prompt, removing extraneous details like roleplay instructions and output format specifications.[55]
- Formatting: Isolates task instructions from contextual narratives, exposing hidden risks.[55]
- Backtranslation: Bypasses direct user obfuscation by feeding the model's initial response into a secondary LLM to infer the prompt that would generate that response.[56, 57] If the target model refuses the reconstructed prompt, the original user request is rejected.[56, 57]

4. DualGuard Verification

DualGuard utilizes a two-stage approach to verify safety.[54] It combines semantic self-evaluation, where the model reflects on its generated response for safety violations, with policy-level verification conducted by a highly accurate external judge model.[54] This approach blocks up to 100% of harmful responses while preserving model usability with low false positive rates on benign queries.[54]

5. Multi-Stage Pipeline Defenses

To optimize security and latency, defenses can be structured as a multi-stage pipeline.[58] Lightweight pre-filters, such as a Linear SVM classifier utilizing text normalization and TF-IDF representations, process incoming traffic.[58] This early-stage filter catches straightforward adversarial prompts with minimal latency, reserving computationally expensive transformer-based monitoring and evaluation for complex, edge-case interactions.[58] These defense strategies demonstrate varying levels of effectiveness depending on the design of the prompt.[59] For instance, providing few-shot demonstrations can produce opposing effects on Role-Oriented Prompts (RoP) and Task-Oriented Prompts (ToP) used for defense.[59] Few-shot examples enhance RoP safety by reinforcing the model's defensive persona but degrade ToP effectiveness by distracting attention away from direct safety guidelines.[59]

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